

Name

- Estimate parameters in the Lotka-Volterra model $\begin{aligned} m' &= \alpha m - \beta m w \\ w' &= -\gamma w + \delta m w \end{aligned}$ for moose (m) and wolves (w).
Plot the predictions of your model.

1. Show some reasoning to estimate birth and death rates for rabbits then write down and solve a simple ODE modeling an initially small population of rabbits $r(t)$ on a lush fertile island. Discuss the results.

■

2.